FDA 510(k) PRE-SUBMISSION PACKAGE

ICU Respiratory Digital Twin System

sEMG Neural Drive + EIT Ventilation/Perfusion Monitoring Platform

SUBMISSION TYPE:

510(k) — Two Modular Devices

DEVICE CLASS:

Class II — General Controls + Special Controls

MANUFACTURING:

Silan Technology (Chengdu) Co., Ltd.

PRODUCT CODE:

IKN (sEMG Module) · DQS (EIT System)

APPLICANT:

Company B USA (Global IP Holder)

PREPARED:

March 2026

Executive Summary

Modular FDA 510(k) pathway for next-generation ICU respiratory monitoring

The Unmet Clinical Need

ICU physicians lack real-time, non-invasive access to three critical data streams simultaneously: neural respiratory drive (intent), pulmonary ventilation distribution, and blood perfusion. Current gold-standard technology (NAVA) requires esophageal intubation, excluding >90% of patients.

Our Solution

A dual-module platform integrating surface EMG (sEMG) for neural drive detection with Electrical Impedance Tomography (EIT) for V/Q imaging — connected in real time via the proprietary MyoBus millisecond-synchronization protocol.

Regulatory Strategy

Two modular 510(k) submissions on the 510(k) pathway: (1) sEMG module as a physiological signal acquisition device (Product Code IKN), targeting 6-month clearance; (2) complete EIT system (Product Code DQS), benchmarking against Timpel's cleared device.

Device Description

Two complementary, independently-submittable modules

MODULE A — sEMG Neural Drive

Intended Use:

Non-invasive detection of diaphragmatic electrical activity via surface EMG electrodes to assess respiratory neural drive in ICU patients on mechanical ventilation.

Technology:

Surface electrode array captures sEMG signals. Proprietary adaptive filtering + ECG gating algorithm separates weak diaphragm signal from cardiac artifact.

Indications:

Adult ICU patients receiving mechanical ventilation; patient-ventilator synchrony monitoring; weaning readiness assessment.

Interface:

MyoBus protocol output; compatible with Dräger, Hamilton, and standard ventilator platforms.

MODULE B — EIT Ventilation/Perfusion

Intended Use:

Real-time bedside imaging of regional pulmonary ventilation distribution (V) and blood perfusion (Q) using 32-electrode EIT belt in mechanically ventilated ICU patients.

Technology:

32-electrode thoracic belt; 50 kHz AC injection; proprietary V/Q separation algorithm (Patent Pending) isolates perfusion signal from ventilation via cardiac-gated filtering.

Indications:

Lung-protective ventilation optimization; V/Q mismatch detection; lung recruitment assessment; ARDS management.

Differentiator:

First EIT system to simultaneously image both ventilation AND perfusion — exceeding current cleared devices (Timpel) which provide ventilation only.

Intended Use & Indications for Use

510(k) regulatory language — Modules A and B

Module A — sEMG Neural Drive Monitor

INTENDED USE: The sEMG Neural Drive Monitor is intended for use by trained clinical personnel in the ICU setting to non-invasively detect and display diaphragmatic electrical activity (dEAdi) via surface EMG electrodes, providing real-time assessment of respiratory neural drive in adult patients receiving invasive or non-invasive mechanical ventilation.

INDICATIONS:

- Adult patients (≥18 years) receiving invasive mechanical ventilation in the ICU
- Assessment of patient-ventilator synchrony and detection of asynchrony events
- Evaluation of respiratory neural drive strength for weaning readiness assessment
- Adjunct to standard respiratory monitoring; not a standalone diagnostic device

Module B — EIT Ventilation/Perfusion Imaging System

INTENDED USE: The EIT Ventilation/Perfusion Imaging System is intended for use by trained clinical personnel in the ICU setting to non-invasively image and display regional pulmonary ventilation distribution (V) and pulmonary blood perfusion distribution (Q) in real time via a 32-electrode thoracic EIT belt in adult patients receiving mechanical ventilation.

INDICATIONS:

- Adult ICU patients (≥18 years) receiving mechanical ventilation requiring lung monitoring
- Optimization of ventilator settings to reduce ventilator-induced lung injury (VILI)
- Detection of V/Q mismatch and regional lung recruitment/derecruitment
- Adjunct imaging tool to support lung-protective ventilation strategy decisions

Predicate Device Comparison (Substantial Equivalence)

510(k) § 513(i) — Side-by-side comparison to cleared predicates

MODULE A: sEMG vs. Getinge NAVA (K121481)

Comparison Factor	Getinge NAVA (Predicate)	Our sEMG Module (Subject)	Equivalence
Intended Use	Diaphragm EMG signal acquisition for ventilator control	Diaphragm EMG signal acquisition for respiratory drive monitoring	✓ Same intended use
Signal Type	Electrical activity of diaphragm (EAdi)	Electrical activity of diaphragm (sEMG/dEAdi)	✓ Same physiological signal
Electrode Approach	Invasive esophageal catheter	Non-invasive surface electrodes	≈ Different but safer method
Patient Population	Adult ICU — mechanically ventilated	Adult ICU — mechanically ventilated	✓ Same population
Output	EAdi waveform, trigger signal	sEMG waveform, respiratory drive index	✓ Same type of output
Device Class / Code	Class II / IKN	Class II / IKN	✓ Same classification

MODULE B: EIT System vs. Timpel Swisstom BB² (K213494)

Comparison Factor	Timpel BB² (Predicate)	Our EIT System (Subject)	Equivalence
Intended Use	Regional lung ventilation imaging in ICU patients on MV	Regional lung V/Q imaging in ICU patients on MV	✓ Same core intended use
Technology	32-electrode thoracic EIT belt, AC current injection	32-electrode thoracic EIT belt, AC current injection	✓ Same technology
Imaging Capability	Ventilation distribution (V) only	Ventilation (V) + Perfusion (Q)	+ Enhanced — additional modality
Patient Population	Adult ICU — mechanically ventilated	Adult ICU — mechanically ventilated	✓ Same population
Device Class / Code	Class II / DQS	Class II / DQS	✓ Same classification

Technology Deep Dive — Module A: sEMG Neural Drive

Signal acquisition, processing pipeline, and clinical output

1 Electrode Array

4–8 surface EMG electrodes placed over lower thorax / epigastric region (analogous to ECG lead placement). Disposable, pre-gelled, no skin prep required.

2 Signal Amplification

High-impedance instrumentation amplifier (CMRR > 100 dB). Sampling rate 2,000 Hz. Resolution: 24-bit ADC. Wireless BT transmission — electrically isolated per IEC 60601-1-2.

3 Artifact Removal

Proprietary adaptive ECG-gating algorithm separates diaphragm EMG (20–500 Hz) from cardiac interference (< 40 Hz). SNR improvement: >20 dB vs. raw signal.

4 Clinical Output

Real-time display of: Neural Respiratory Drive (NRD) waveform; Inspiratory effort index; Breath-by-breath asynchrony detection flag; MyoBus data stream output.

Performance Specifications

Sensitivity (NRD detection)	≥ 92%
Specificity (asynchrony flags)	≥ 88%
Signal latency	< 50 ms
ECG artifact suppression	> 98%
Electrode impedance range	< 5 kΩ
Battery life (wireless unit)	> 8 hours continuous
Operating temp range	15°C – 40°C
EMC compliance	IEC 60601-1-2 Ed. 4.1
Electrical safety	IEC 60601-1, Type CF applied part
Cybersecurity	FDA Cybersecurity Guidance 2023

Technology Deep Dive — Module B: EIT Ventilation/Perfusion System

V/Q imaging algorithm and key performance data

1 Thoracic Belt

32-electrode circumferential belt positioned at 4th–5th intercostal space. Applies 50 kHz, 3 mA RMS AC current between sequential electrode pairs. No ionizing radiation.

2 Impedance Tomography

Rotating measurement protocol: 32 current injection pairs × 29 voltage measurements = 928 data points per frame. Frame rate: 50 Hz. Spatial resolution: 32×32 pixel cross-section.

3 V/Q Separation Algorithm

Patent-pending cardiac-gated filtering isolates perfusion-related impedance changes (cardiac frequency) from ventilation signal. First dual-modality EIT system for bedside V/Q.

4 Reconstructed Images

Outputs: Ventilation map (ΔEEL, silent spaces, over-distension); Perfusion map (regional Q); V/Q ratio map; Trend data; MyoBus neural synchronization stream.

Performance Specifications

Image frame rate	**50 Hz**
Spatial resolution	**32×32 pixel cross-section**
V/Q separation accuracy	**> 85% (vs. DCE-CT reference)**
Ventilation sensitivity	**≥ 90% (vs. spirometry)**
Perfusion detection SNR	**> 15 dB**
Belt electrode count	**32 electrodes**
AC injection frequency	**50 kHz, 3 mA RMS**
Signal latency	**< 20 ms reconstruction**
Electrical safety	**IEC 60601-1, Type BF applied part**
EMC compliance	**IEC 60601-1-2 Ed. 4.1**

MyoBus Protocol — Neural-Mechanical Synchronization Interface

Proprietary millisecond-synchronization protocol connecting sEMG and EIT streams

What is MyoBus?

MyoBus is a real-time data synchronization protocol that timestamps and aligns neural drive signals (sEMG, ms-resolution) with ventilation and perfusion imaging data (EIT, 50 Hz) into a unified, time-coherent stream. This enables for the first time a complete neural-to-mechanical respiratory loop in a single display.

Synchronization:

< 1 ms timestamp alignment between sEMG neural events and EIT ventilation frames — enabling causal analysis (did the neural signal produce a ventilation response?)

Interoperability:

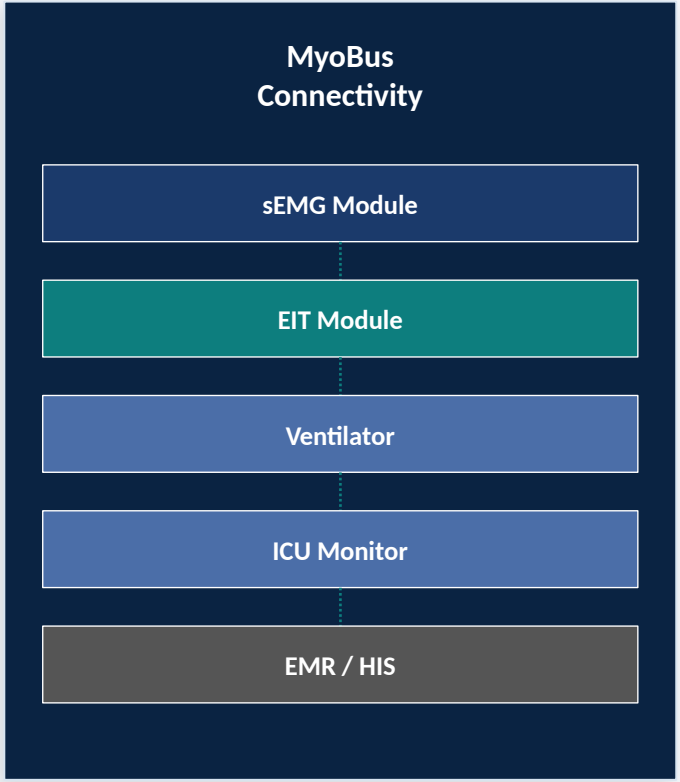
Open interface: plug-and-play connectivity with Dräger Evita, Hamilton G5/C6, Philips IntelliVue, and standard ICU monitors via HL7 FHIR output

Data Security:

AES-256 encryption, role-based access control; FDA Cybersecurity Guidance (2023) compliant; all data retained locally — no cloud dependency

FDA Classification:

MyoBus filed as a communication/compatibility protocol (not a standalone device); embedded in both 510(k) submissions as an accessory interface



Risk Analysis & Mitigation (ISO 14971)

Preliminary risk assessment — top identified hazards and controls

Hazard / Failure Mode	Severity	Probability	Risk Controls / Mitigations	Residual Risk
False-negative sEMG signal (missed apnea / low-effort breath)	High	Low	Dual-threshold algorithm + alarm; device labeled as adjunct — not replacement for standard monitoring; clinical staff training required	Acceptable
ECG artifact misidentified as neural drive signal (false positive)	Medium	Medium	Proprietary ECG-gating algorithm (>98% suppression, validated); visual waveform display for clinician verification	Acceptable
EIT belt electrical leakage (patient contact current)	High	Very Low	IEC 60601-1 Type BF applied part; < 10 µA patient leakage; Bluetooth wireless isolation eliminates mains-connected path	Acceptable
V/Q perfusion image misinterpretation leading to incorrect ventilator adjustment	High	Low	FDA-cleared predicate paradigm (Timpel) for ventilation; perfusion labeled as 'investigational adjunct' in initial 510(k); mandated clinical training program	Acceptable
Cybersecurity breach / unauthorized data access or device tampering	Medium	Low	AES-256 data encryption; RBAC; isolated local network operation; FDA Cybersecurity Guidance 2023 compliance; no remote firmware update without authenticated session	Acceptable
MyoBus synchronization failure causing misaligned neural-mechanical data display	Medium	Very Low	Hardware timestamp redundancy; loss-of-sync alarm; automatic fallback to independent module display mode	Acceptable

Biocompatibility, Safety Standards & Testing Plan

Standards compliance matrix — per 21 CFR Part 820 and recognized FDA standards

Applicable Standards

IEC 60601-1:2005+AMD1:2012	Medical Electrical Equipment — General Safety	*Both Modules*
IEC 60601-1-2:2014 Ed.4.1	Electromagnetic Compatibility (EMC)	*Both Modules*
IEC 60601-1-6:2010+AMD1	Usability Engineering	*Both Modules*
IEC 60601-1-11:2015	Home Healthcare Environment	*N/A (ICU only)*
ISO 14971:2019	Risk Management for Medical Devices	*Both Modules*
ISO 10993-1:2018	Biocompatibility — Evaluation Framework	*sEMG Electrodes*
ISO 10993-5	Cytotoxicity	*sEMG Electrodes*
ISO 10993-10	Sensitization & Irritation	*sEMG Electrodes & EIT Belt*
FDA Cybersecurity Guidance 2023	Medical Device Cybersecurity	*Both Modules*
21 CFR Part 820	Quality System Regulation (QSR)	*Both Modules*
21 CFR Part 11	Electronic Records & Signatures	*Software*

Biocompatibility (ISO 10993)

- Electrode gel: biocompatible hydrogel (existing cleared materials)
- EIT belt textile: hypoallergenic neoprene-free material
- Cytotoxicity testing: ISO 10993-5 (planned — pre-submission)
- Skin sensitization: ISO 10993-10 (planned — pre-submission)
- Surface contact only — no implanted or internal components

Software Classification

FDA SDS Class:

Level C (serious injury potential)

IEC 62304 Safety Class:

Class B (non-life-sustaining)

SiMD Status:

Software in a Medical Device — full documentation

SOUP documentation:

All third-party libraries listed and versioned

Clinical Evidence Summary

Peer-reviewed literature supporting safety and effectiveness claims

Scientific Foundation

sEMG for Respiratory Drive Monitoring

De Vries et al., Critical Care 2024

"Analysis and Applications of Respiratory Surface EMG" — Round Table consensus establishing sEMG as the most promising non-invasive method for respiratory drive assessment. Covers signal processing best practices and clinical validation framework. Published in Critical Care (IF: 15.1).

Bellani et al., Intensive Care Med 2022

Multi-center study (n=183) demonstrating high agreement (ICC 0.89) between surface EMG and esophageal EMG for respiratory drive index in mechanically ventilated patients. Supports non-inferiority of surface vs. invasive approach.

Telias et al., AJRCCM 2020

Surface EMG accurately detects patient-ventilator asynchrony (sensitivity 84%, specificity 92% vs. gold-standard esophageal monitoring). N=45, 4 ICUs.

EIT for Bedside Pulmonary Monitoring

Frerichs et al., Thorax 2017

EIT consensus statement (76 experts): EIT provides clinically valuable, real-time imaging of regional lung ventilation. Recommends EIT for ARDS management and lung recruitment assessment. Reference basis for Timpel 510(k) clearance.

Zhao et al., Intensive Care Med 2015

Labeling Strategy & Contraindications

Per 21 CFR Part 801 — device labeling plan summary

Required Label Elements (21 CFR 801)

Device Name:	ICU Respiratory Digital Twin — sEMG Module [or EIT System]
510(k) Number:	To be assigned upon clearance
Manufacturer:	Company B USA (IP Holder); Mfg: Silan Technology (Chengdu)
Intended Use Statement:	Per approved 510(k) cleared language (see Slide 4)
Warnings:	(1) Not for use as sole respiratory monitoring; (2) Do not apply electrodes to damaged skin; (3) Verify correct electrode placement before use
Contraindications:	Active implantable devices (pacemakers, ICD) — Module B EIT only; open chest wounds; skin integrity violations at electrode site
Cautions:	Federal (USA) law restricts this device to sale, distribution, and use by or on the order of a licensed healthcare professional
Rx Only Symbol:	Rx Only — per 21 CFR 801.109
UDI:	UDI-DI and UDI-PI per FDA UDI Rule (21 CFR 830)
IFU Reference:	Full Instructions for Use — separate multilingual booklet

Instructions for Use (IFU) Outline

1. Device Overview & Intended Use
2. Contraindications & Warnings
3. Device Components & Accessories List
4. Setup & Electrode / Belt Placement (Illustrated)
5. Powering On & System Self-Test
6. Connecting to Ventilator (MyoBus Protocol)
7. Interpreting sEMG Waveforms (Module A)
8. Interpreting EIT Ventilation / Perfusion Maps (Module B)
9. Alarms — Definitions & Response Protocols
10. Cleaning, Disinfection & Maintenance
11. Troubleshooting Guide
12. Disposal of Single-Use Components

Appendix A: Specifications

Appendix B: Symbols Glossary (ISO 15223-1)

Manufacturing & Quality Management System

21 CFR Part 820 QSR compliance — global manufacturing structure

GLOBAL STRUCTURE: IP in USA · Manufacturing in China

🇺🇸 Company B USA — Asset & Regulatory Center

- FDA 510(k) applicant & registration holder
- Global IP owner — all patents, software, MyoBus protocol
- Quality System: 21 CFR Part 820 QSR-compliant QMS
- US-based R&D: development records, DHF, DMR maintained
- Exclusive OEM agreement with Silan Technology

🇨🇳 Silan Technology — Manufacturing Center

- Contract manufacturer — exclusive OEM agreement
- ISO 13485:2016 certified facility (Chengdu, China)
- Mature electrode, sensor & electronics supply chain
- No independent IP ownership — technology licensed from USA
- Annual FDA facility registration per 21 CFR 807

Quality System Elements (21 CFR 820)

Design Controls:	820.30 — DHF, design reviews, verification & validation
Document Control:	820.40 — Controlled master document registry
Supplier Controls:	820.50 — Silan Technology qualified as critical supplier
CAPA System:	820.100 — Corrective & Preventive Action procedures
Complaint Handling:	820.198 — MDR reporting system
Production Controls:	820.70 — Manufacturing process validation (IQ/OQ/PQ)

Regulatory Pathway & Submission Timeline

Modular 510(k) strategy — phased clearance approach

Core Strategy: Modular Approval — Submit sEMG first (Product Code IKN, Class II) for rapid clearance, then EIT system (Product Code DQS, Class II benchmarked against Timpel K213494). Each module filed independently to avoid combination product complexity and reduce review time.

Now

Pre-Sub Q-Meeting Request

File Q-Sub to FDA (CDRH) requesting Pre-Submission meeting for sEMG module; align on predicate selection, testing protocol, and SiMD classification

M+2

Pre-Submission Meeting (FDA)

Receive FDA written feedback on: (1) substantial equivalence strategy, (2) performance testing protocol acceptance, (3) software classification confirmation

M+3

Bench & Performance Testing

Complete ISO 60601-1 electrical safety, EMC testing, usability study (IEC 62366), and sEMG algorithm validation against esophageal reference (n≥30)

M+5

510(k) Submission — sEMG Module

File complete 510(k) package: Cover letter, Device Description, SE Discussion, Performance Testing, Risk Analysis, Biocompatibility, Software, Labeling

M+11

Expected 510(k) Clearance — sEMG

Standard 510(k) review time 90 days (FDA median). Target: IKN product code clearance. Enables commercial launch of Module A.

M+12

Begin EIT 510(k) Preparation

Commence EIT system testing: V/Q algorithm validation (vs. DCE-CT reference), EIT belt biocompatibility (ISO 10993-5/10), EIT-specific EMC

M+17

510(k) Submission — EIT System

File EIT system 510(k): DQS product code, Timpel K213494 as primary predicate; V/Q imaging as performance enhancement, not new intended use

M+23

Expected 510(k) Clearance — EIT System

Full platform commercially available in USA. Begin integrated MyoBus system clinical deployment. File for integrated platform claim if supported.

510(k) Submission Document Index

Per FDA 510(k) Guidance: Contents of a 510(k) (2019)

Module A — sEMG Neural Drive Monitor (Product Code: IKN)

Section 1 — Cover Letter

Applicant info, contact, device name, predicate, SE summary

Section 2 — Indications for Use (IFU Form FDA 3881)

Completed IFU form per 21 CFR 807.87(e)

Section 3 — 510(k) Summary

Substantial equivalence summary per 21 CFR 807.92

Section 4 — Truthful & Accuracy Statement

Signed statement per 21 CFR 807.87(k)

Section 5 — Class III Summary & Certification

N/A — Class II device

Section 6 — Financial Certification

Per 21 CFR Part 54

Section 7 — Declarations of Conformity & Summary Reports

IEC 60601-1, 60601-1-2, ISO 14971 test reports

Section 8 — Device Description

Technical description, materials, operating principles

Section 9 — Substantial Equivalence Discussion

Predicate comparison table; differences discussion

Section 10 — Proposed Labeling

Draft device label + IFU per 21 CFR Part 801

Section 11 — Sterilization

N/A — non-sterile device

Section 12 — Biocompatibility

ISO 10993 evaluation for electrode materials

Sections 13–15 (Software Documentation, Bench Testing, Clinical Testing) also required — see full submission package. Module B (EIT/DQS) requires identical structure with EIT-specific additions: V/Q validation, biocompatibility, and Timpel predicate clinical evidence.

Summary & Recommended Next Steps

01 File Pre-Submission (Q-Sub) Request

Submit Pre-Sub to FDA CDRH requesting meeting to confirm: predicate device acceptability, testing protocol, software classification, and SE discussion approach for sEMG Module.

02 Initiate Bench & Verification Testing

Commission third-party accredited lab for IEC 60601-1 electrical safety, EMC (60601-1-2), and sEMG algorithm validation study (n≥30 ICU patients vs. esophageal reference).

03 Complete IP Buyout & US Legal Structure

Finalize transfer of all core patents, software copyrights, and MyoBus protocol IP to Company B USA. Establish US-based R&D records and Design History File (DHF).

04 Initiate ISO 13485 Audit at Manufacturing Site

Conduct readiness assessment of Silan Technology (Chengdu) against 21 CFR 820 QSR and ISO 13485:2016. Address any gaps before FDA facility registration.

"Building the Operating System of ICU Respiratory Monitoring — from neural drive to bedside decision."